Asymptotics 渐近

2种误差：① 计算误差 ($M \to \infty$)
② 数据误差 ($n \to \infty$)

Outline

Topics

- The notion of asymptotic analysis.
- The two asymptotic regimes considered in this class.

Rationale

In statistics, asymptotic analysis is an important tool to understand any estimator, including Bayesian estimators and Monte Carlo methods.

A first type of asymptotics: consistency of Monte Carlo methods

Example: suppose you are trying to estimate the slope parameter in last week's regression model.

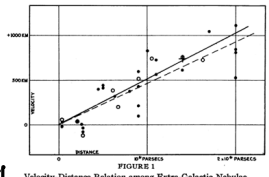


FIGURE 1
Velocity-Distance Relation among Extra-Galactic Nebulae.

slope, sd
unknown parameter

true

- Typical situation:
  - Fix one Bayesian model and one dataset.
  - You are interested in $g^* = \mathbb{E}[g(X)|Y=y]$ $g(x)$：目标函数 posterior: $P(X|Y=y)$
    - e.g., $g(\text{slope}, \text{sd}) = \text{slope}$ in last week's regression example.
  - You try simPPLe with $M = 100$ iterations, get a Monte Carlo approximation $\hat{G}_M \approx \mathbb{E}[g(X)|Y=y]$
  - Then you try $M = 1000$ iterations to gauge the quality of the Monte Carlo approximation $\hat{G}_M$.
  - Then you try $M = 10000$, etc.
- Mathematical guarantee: we proved (Monte Carlo) consistency,
  - i.e. that $\hat{G}_M$ can get arbitrarily close to g^*.

Monte Carlo consistency: the limit $M \to \infty$ gives us a first example of asymptotic regime: "infinite computation." 是在计算次数M下的渐近

$\hat{g}_M \xrightarrow{M \to \infty} g^*$

在#computation个的情况下渐近.

A second type of asymptotics: "big data" N大

M：computational budget
n：data size.

Question: even after $M \to \infty$, can there still be an error in the "infinite compute" limit g^*? Yes, because of model mis-specification and limited data.

You will explore this question in this week's exercises.

Optimality

Outline

Topics

- Average loss optimality of Bayes estimators
- Limitations of this notion of optimality.

Rationale

Formalize and justify one of the motivations for Bayesian methods covered in the first lecture.

Setup

We go back to the decision theoretic setup from week 2.

Estimator

Definition: an estimator is a function δ such that:

input: y

output: Action.

- δ takes a dataset y as input (i.e. the observations), $\delta(y)$, and
- returns an action, i.e., $\delta(y) \in A$.

data → action.

$\delta: y \to A$

Examples:

① Bayes estimator

② MLE

- the Bayes estimator,
- maximum likelihood estimator (when the action is a parameter you are trying to estimate).

Optimality: motivation

Consider the following situation—you are watching a talk, where the presenter explains they did the following:

- Let:
  - δ_{B} denote the Bayes estimator, and
  - δ_{o}, another estimator, say based on a sophisticated deep neural network.
- To compare the two estimators, the presenter generated a large number of synthetic datasets $(X^{(m)}, Y^{(m)})$ by forward simulation from the model.
- They evaluated the performance of the Bayes estimator as follows:

- apply the estimator: $a^{(m)} = \delta_{\mathrm{B}}(Y^{(m)})$, → truth.
- compute the loss: $l^{(m)} = L(a^{(m)}, X^{(m)})$, and → Action
- return the average loss as the measure of performance:

$$\underline{\text{average loss}} = \frac{1}{M}\sum l^{(m)}.$$

会因为M小而造成 Monte carlo Approx. 有误差

- Then, they did the same with the other estimator, δ_{o}.
- Suppose that the presenter reported a lower average loss for the other estimator, δ_{o}

Poll: Which of these statements is true?

1. The deep neural network can perform better because it involves more parameters (over-parameterization)
2. The deep neural network can perform better because of a combination of lucky initialization and noisy gradients
3. The deep neural network can perform better because of some other reason
4. This cannot happen unless the loss function is non-convex
5. This cannot happen because of some other reason Bayes estimator : minimize Avg Loss

Optimality of Bayesian estimators: theorem

sample.

Theorem: let δ_{B} denote the Bayes estimator, and δ_{o}, any other estimator. Then:

$$\mathbb{E}[L(\delta_{\mathrm{B}}(Y), X)] \leq \mathbb{E}[L(\delta_{\mathrm{o}}(Y), X)]. \tag{1}$$

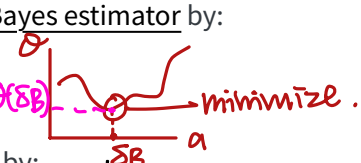
Proof: by the law of total expectation, we can rewrite the left hand side of Equation 1 as:

$$\mathbb{E}[L(\delta_{\mathrm{B}}(Y), X)] = \mathbb{E}[\mathbb{E}[L(\delta_{\mathrm{B}}(Y), X)|Y]]. \tag{2}$$

We will start by rewriting the part in red.

To do so, denote the objective being minimized in the Bayes estimator by:

定义 posterior expected loss: $\mathcal{O}(a) = \mathbb{E}[L(a, X)|Y]$



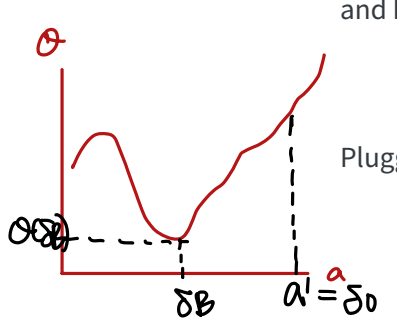
Note: by definition, we can rewrite our Bayes estimator by:

$$\delta_{\mathrm{B}} = \arg\min_a \mathcal{O}(a),$$

对每个Y, Bayes estimator都选 Expected loss 最小的那个.

and hence:

$$\mathbb{E}[L(\delta_{\mathrm{B}}(Y), X)|Y] = \mathcal{O}(\delta_{\mathrm{B}}) = \mathcal{O}(\arg\min_a \mathcal{O}(a)). \tag{3}$$



Plugging-in Equation 3 into Equation 2, we get:

$$\mathbb{E}[\mathbb{E}[L(\delta_{\mathrm{B}}(Y), X)|Y]] = \mathbb{E}[\mathcal{O}(\arg\min_a \mathcal{O}(a))].$$

Now note that for all $a' \in A, \mathcal{O}(\arg\min_a \mathcal{O}(a)) \leq \mathcal{O}(a')$, in particular, this is true for $a' = \delta_{\mathrm{o}}(y)$, hence:

$$\begin{aligned}\mathbb{E}[L(\delta_{\mathrm{B}}(Y), X)] &= \mathbb{E}[\mathcal{O}(\arg\min_a \mathcal{O}(a))] \\ &\leq \mathbb{E}[\mathcal{O}(\delta_{\mathrm{o}}(Y))] \\ &= \mathbb{E}[L(\delta_{\mathrm{o}}(Y), X)].\end{aligned}$$

Discussion: Why do you think the presenter observed $\mathbb{E}[L(\delta_{\mathrm{B}}(Y), X)] > \mathbb{E}[L(\delta_{\mathrm{o}}(Y), X)]$ in their experiment?

① Monte carlo误差：M太小

② Another estimator：没找到最优.

Bayes estimator: Minimize expected avg. loss under joint distribution.

Given $E[L(\delta_B(Y), X)] = E[E[L(\delta_B(Y), X)]]$. why $E[L(\delta_B(Y), X)] \leq E[L(\delta_o(Y), X)]$.

$O_a = E[L(a, X) | Y]$.

$\delta_B(Y) = \arg\min_a O(a) \Rightarrow O(\delta_B(Y)) \leq O(\delta_o(Y))$

$E[O(\delta_B(Y))] \leq E[O(\delta_o(Y))]$

Decision theoretic point estimation

Outline

Topics

- Deriving a point estimate from decision theory.

Rationale

We have seen in week 2 some examples of point estimates (posterior mean, posterior mode).

These are actually special cases of decision theory with specific choices of loss functions.

This page provides a general framework to answer the question: "how to summarize a posterior distribution with one point?"

Setup

The decision theoretic setup from week 2.

$\delta_B(Y) = \arg\min_{a \in A} E[L(a,X)|Y]$.

Example

X: unknown param. Y: observed data A: action.

⇒ 看到 Y 后选 1 个 $a \in A$.

Assume: a *square loss*, $L(a,p) = (a-p)^2$, where $a \in A = \mathbb{R}$.

Some initial simplification: on the objective function…

$$\begin{aligned}
\delta_{\mathrm{B}}(Y) &= \arg\min\{\mathbb{E}[L(a,X)|Y] : a \in A\} \\
\text{开平方} &= \arg\min\{\mathbb{E}[(X-a)^2|Y] : a \in A\} \\
\text{展开平方} &= \arg\min\{\mathbb{E}[X^2|Y] - 2a\mathbb{E}[X|Y]] + a^2 : a \in A\} \\
&= \arg\min\{-2a\mathbb{E}[X|Y]] + a^2 : a \in A\}
\end{aligned}$$

在看到 data Y 后，选那个让 posterior expected loss 最小的 Action.

or 因为对 a 取 argmin，这一项无关 a.

Question: under a square loss, δ_{B} can be simplified to…

☆看 loss function ⇒ 看 posterior 的什么.

squared loss = posterior mean

Absolute loss = posterior median

0-1 loss = posterior mode.

$\min_a a^2 - 2a\mu$, $\mu = E[X|Y]$

sol: $\frac{d}{da}(a^2 - 2a\mu) = 2a - 2\mu = 0$

$\Rightarrow a = \mu$.

posterior Mean.

$\Rightarrow \delta_B(Y) = E[X|Y] = \int x f(x|Y)dx$.

X: latent

Y: observed

squared loss: posterior mean.

Decision theoretic set estimation

Outline

Topics

- Deriving a set estimate from decision theory.

Rationale

We have seen in week 2 some examples of set estimates (quantile based, highest density set).

These are actually special cases of decision theory with specific choices of loss functions.

This page provides a general framework to answer the question: "how to summarize a posterior distribution with one **set**?"

Context

- The weakness of point estimates is that they do not capture the uncertainty around the value.
- **Idea:** instead of returning a single point, return a set of points
  - usually an interval,
  - but this can be generalized
- Bayesian terminology: **credible interval** ($\neq$ frequentist *confidence intervals*)
- Goals:
  - We would like the credible interval to contain a fixed fraction of the posterior mass (e.g. 95%)
  - At the same time, we would like this credible interval to be as short as possible given that posterior mass constraint

Bayes estimator formalization

用decision theory ⇒ 区间 credible interval.

Pick:

- $A = \{[c, d] : c < d\}$, 每个action=选择1个区间[c,d].
- consider the loss function given by

  如果真值被covered, loss=1.

$$L([c, d], x) = \mathbb{1}\{x \notin [c, d]\} + k(d - c)$$

  K>0, 区间越长, 惩罚越大.

  for some tuning parameter k to be determined later.

HPD interval

We get:

$$
\begin{aligned}
\delta_{\mathrm{B}}(Y) &= \arg\min\{\mathbb{E}[L(a,X)|Y] : a \in A\} \\
&= \arg\min\{\Pr[X \notin [c,d]|Y] + k(d-c) : [c,d] \in A\} \\
&= \arg\min\{\Pr[X < c|Y] + \Pr[X > d|Y] + k(d-c) : [c,d] \in A\} \\
&= \arg\min\{\Pr[X \leq c|Y] - \Pr[X \leq d|Y] + k(d-c) : [c,d] \in A\}
\end{aligned}
$$

引入LOSS
解释indicator.

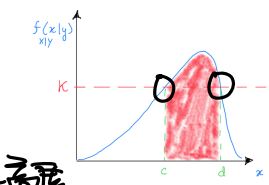
Here we assume the posterior has a continuous density f to change $<$ into $\leq$.

As we did with point estimation, we take the derivative with respect to c and set to zero (then will do the same thing for d).

Notice that $\Pr[X \leq c|Y]$ is the posterior cumulative distribution function (CDF)[1], so taking the derivative with respect to c yields a density:[2]

$$f_{X|Y}(c) - k = 0,$$

$f_{X|Y}(c) = k.$
$f_{X|Y}(d) = k.$



so we see the optimum will be the smallest interval $[c, d]$ such that

$$f(c) = f(d) = k.$$

区间的2端都落在post. density的同一高度.

Finally, set k to capture say 95% of the mass (area in red in the figure shows a 50% example, we would lower k to get to 95%).

Footnotes

1. Recall that a CDF is defined as $F(x) = \mathbb{P}(X \leq x)$. So a posterior CDF is just $\mathbb{P}(X \leq x|Y)$. ↩
2. Let us review the argument why "taking the derivative … yields a density". Let f denote a density and F, the CDF of the same random variable. From the definition of a density, $F(x) = \int_{-\infty}^{x} f(x')\mathrm{d}x'$ (i.e., take $A = [-\infty, x]$ in the definition of a density). Hence by the Fundamental theorem of calculus (i.e. the fact that an integral is an "anti-derivative"), we have that

$$f(x) = \frac{\mathrm{d}F(x)}{\mathrm{d}x},$$

when the derivative exists. ↩

Calibration 校准.

Outline

Topics

- Notion of calibration.
- Nominal versus actual coverage.

Rationale

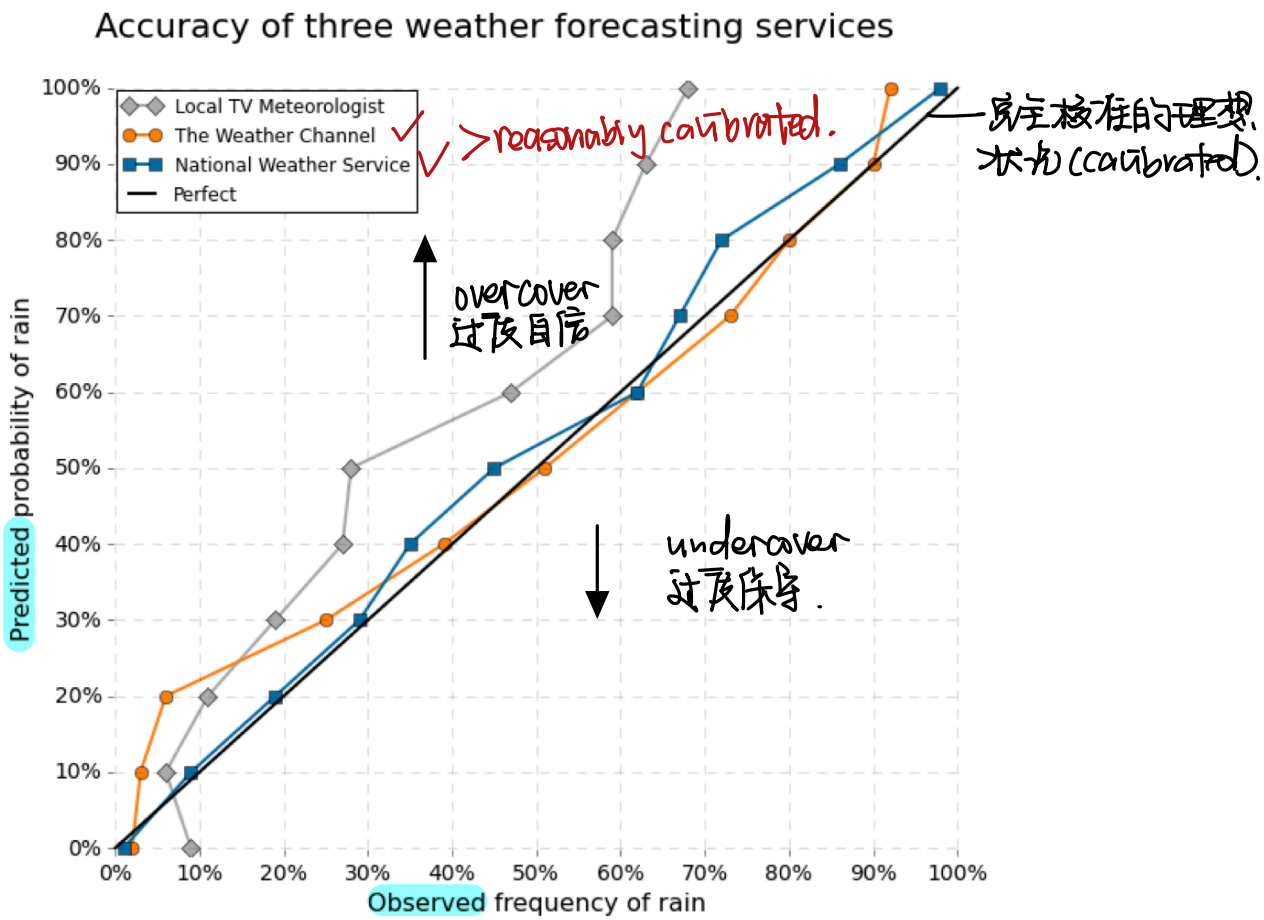
Returning a credible region is helpful as it conveys how much certain or uncertain we are about our answer.

However, these estimates of uncertainty can be misleading if they are too confident or too hesitant. The notion of calibration formalizes this intuition.

Example

- Familiar in Vancouver: “Tomorrow: 90% chance of rain,”
  - here, “90%” is an example of nominal coverage.
- Of those days where the forecast says “Tomorrow: 90% chance of rain”…
  - what **fraction of those days** did it actually rain?
  - That fraction is an example of **actual coverage.** 真的下雨比例.

Definition: a measure of uncertainty is **calibrated** if its nominal coverage matches its actual coverage.



Question: which weather source(s), if any, provide reasonably calibrated rain uncertainty estimates?

optimality ≠ calibration. Bayes estimator 在 expected loss 下最优
但不保证 calibration.

Bayesian calibration: well-specified case

Outline

Topics

- Bayesian calibration.
- Guarantees when the model is well-specified.

Rationale

We introduced in the previous page the important notion of calibration.

The question we consider now: **are credible intervals calibrated?** nominal cov = actual cov.

We will give a two-part answer: first, answering the above question for well-specified models (this page), then, for mis-specified models (next page).

Bayesian calibration

Thought experiment: what if a certain Bayesian inference method was used many times?

- For example, different labs, each studying similar but different datasets,
- i.e., they are replicating an "experiment."
- This leads to a *frequentist analysis of Bayesian procedures*—a very useful thing to do!

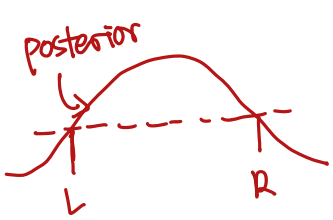
☆ **Calibration of credible intervals**: a 90% credible interval is **calibrated** if it contains the true parameter **90% of the time**.

What do we mean by "true parameter"?

真实data

- Start with a joint distribution $p^\star(x, y)$, which we will call "nature."
- For each "experiment": use $p^\star$ to simulate both a true parameter $x^\star$ → true param. and an associated dataset $y^\star$: $(x^\star, y^\star) \sim p^\star$. (= perform forward simulation).

true data



What do we mean by "90% credible interval"? A function $C(y) = [L(y), R(y)]$ which:

1. computes the posterior $\pi(\cdot|y)$, and
2. selects left and right end points, l, r such that $\int_l^r \pi(x|y)\mathrm{d}x = 0.9$. nominal coverage.

Recall an examples of credible interval we discussed: quantile-based intervals.

What do we mean by "90% of the time"?

- Loop over $1, 2, \ldots,$ numberOfExperiments 重复实验.
  - generate synthetic data $(x^\star, y^\star) \sim p^\star$
  - compute the credible $C(y^\star)$
  - record if the true parameter is in the interval, i.e. if $x^\star \in C(y^\star)$

目标: $P^*(X \in C(Y)) = 0.9$

- Consider the limit, as numberOfExperiments $\to \infty$, of the fraction of times the true parameter is in the interval $\frac{1}{N}\sum 1\{x^* \in C(y^*)\}$.
  - If the limit is equal to 0.9 we say the credible interval is calibrated (great!)
  - If the limit is close to 0.9 we say the credible interval is approximately calibrated (that's not too bad)
  - If the limit is higher than 0.9, we say the credible interval is over-conservative (that's not too bad)
  - If the limit is lower than 0.9, we say the credible interval is anti-conservative (bad!)

Well-specified vs mis-specified models

- There are two joint distributions involved in the thought experiment
  - p^* used to generate data
  - p, used internally by the credible interval procedure to define a posterior $\pi(x|y) \propto p(x, y)$
- We can consider the following setups
  - Well-specified setup $p^* = p$
  - Mis-specified, $p^* \neq p$
- For now: focus on the well-specified setup.

Numerical exploration

Using the discrete model for Delta launches from Exercise 2, we will now implement and run the algorithm described above under "What do we mean by 90% of the time?"…

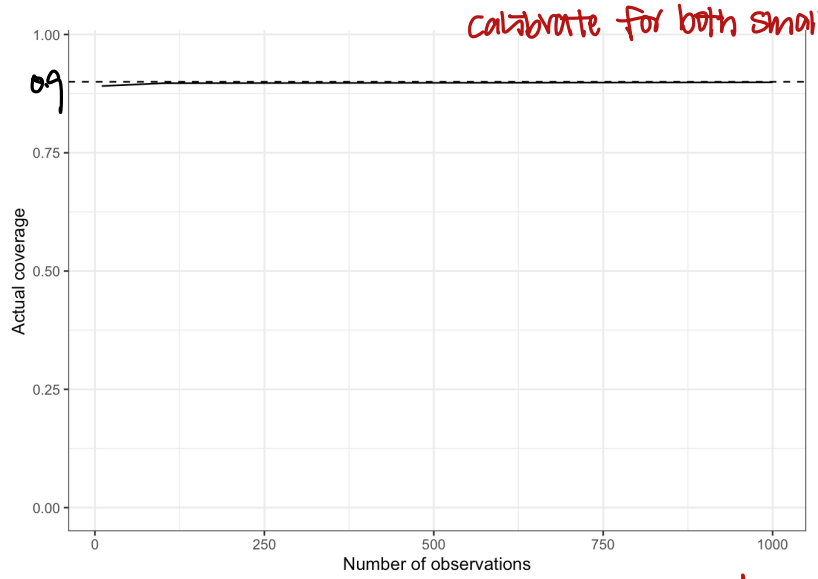
What do you expect?

1. calibrated for small data, calibrated for large data
2. not calibrated for small data, calibrated for large data
3. only approximately calibrated for both small and large data
4. none of the above

▶ Code

calibrate for both small dataset and large dataset.

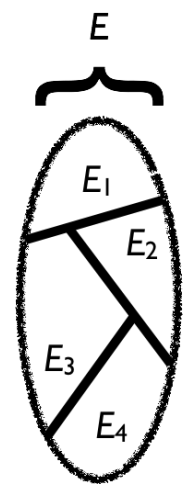
随着n↑, Actual cov 稳定在0.9附近.



Contrast with frequentist intervals, credible interval can be calibrated under small dataset.
↳ need large dataset (CLT)

Mathematical underpinnings

- Denote the data by Y.
- Parameter of interest: X.
- Denote the credible interval obtained from y by $C(y)$.
- We want to show $\mathbb{P}^*(X \in C(Y)) = 0.9$.
- By construction, the interval $C(y)$ satisfies, for all observed data y, $\mathbb{P}(X \in C(y) | Y = y) = 0.9$.



- Recall the law of total probability:
  - for any partition E_i of the sample space,
  - we have $\mathbb{P}(A) = \sum_i \mathbb{P}(A, E_i)$.
- Here: $A = (X \in C(Y)), E_i = (Y = y_i)$
  - where $\{y_i\}$ denotes the possible values the data can potentially take across all possible simulated experiments.
- Putting it all together and applying chain rule:

$$
\begin{aligned}
\mathbb{P}^*(X \in C(Y)) &= \sum_i \mathbb{P}^*(X \in C(Y), Y = y_i) \\
&= \sum_i \mathbb{P}^*(Y = y_i)\mathbb{P}^*(X \in C(y_i) | Y = y_i) \\
&= \sum_i \mathbb{P}^*(Y = y_i)\underbrace{\mathbb{P}(X \in C(y_i) | Y = y_i)}_{0.9} \qquad \text{(assuming model is well-specified} \\
&= \sum_i \mathbb{P}^*(Y = y_i)0.9 \\
&= 0.9 \sum_i \mathbb{P}^*(Y = y_i) = 0.9.
\end{aligned}
$$

P=P*

if Bayesian model is well-specified ⇒ Bayesian credible interval are calibrated.
↳ 不用asymptotic, CLT, large N.

Bayesian calibration: mis-specified case

Outline

Topics

- Guarantees when the model is mis-specified.
- The Bernstein-von Mises (**BvM**) theorem.

Rationale

Model mis-specification is the norm. What happens to calibration in that case?

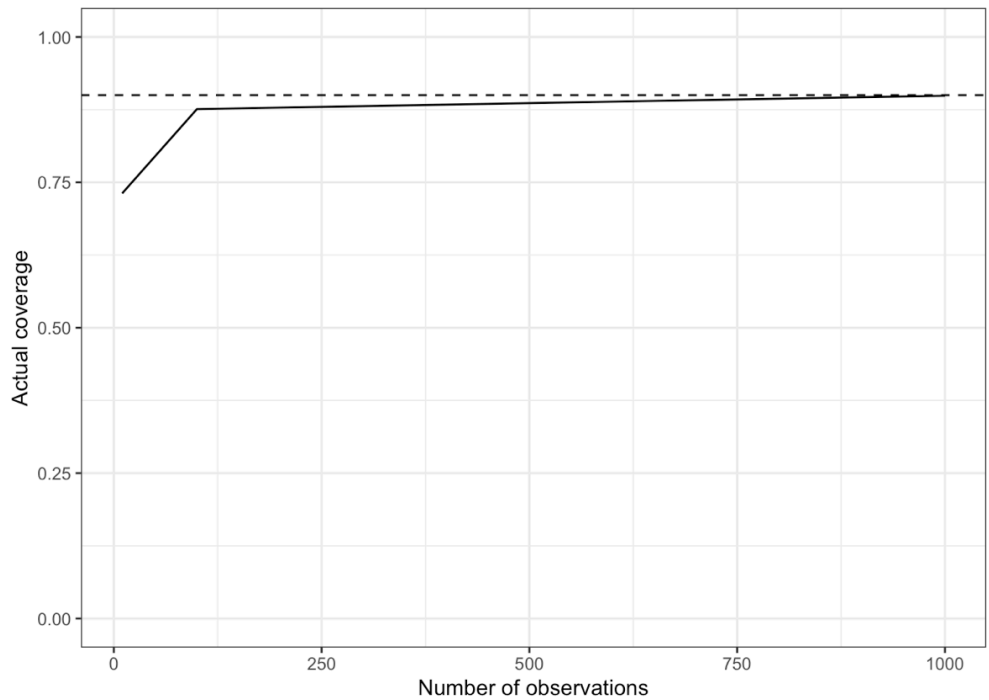
Misspecified models and calibration

What to do if your model is misspecified?

One obvious possibility is to make the model better. But in the following we show that under certain conditions, another possibility is to collect more data.

- Now suppose we are using the "wrong prior", i.e.:
  - data generation uses uniform prior…
  - …but we base our posterior computation on a different, non-uniform prior.
- Similarly to the last page, let's do it for small (10 launches), medium (100), and large datasets (1000), plotting the **nominal coverage** (dashed) against the **actual coverage** (solid line)

▶ Code



- Bad news: for small datasets we are no longer calibrated, in the worst possible way
  - Higher than the dashed line: would have meant inference is being **conservative**, i.e. more right than it actually claimed. That's not too bad.
  - Lower than the dashed line: we are being **overconfident** or **anti-conservative**, which in some cases can be reckless
- Good news: in this case, the problem gets quickly corrected as the dataset gets larger. **Why?**

Asymptotic calibration under misspecification of the prior

- There is no general calibration theory for misspecified models, only special cases (outside of that, use simulation studies or cross-validation techniques)
- An example where we do have some theory: when the data points y_i are iid given the parameters x.

Proposition: under certain conditions (outlined below), even when the prior is misspecified, the actual coverage of credible intervals converges to the nominal coverage. calibrated.

Outline:

- the above proposition is a consequence of a stronger result: **Bernstein–von Mises theorem** (BvM).
- BvM theorems provide conditions where the posterior converges to a normal as the data size increases. calibrated if $n \to \infty$
  - Crucially, the parameters of the limiting normal do not depend on the

prior!
    - As a result, when BvM holds, prior misspecification is eventually washed out by the likelihood.

BvM Theorem

Setup:

- Assume there is a single true x^*.
- Generate datasets of increasing size n,
    - $Y_{1:2}, Y_{1:3}, \ldots,$
    - $Y_{1:n} = (Y_1, \ldots, Y_n),$ iid
    - $Y_i \sim p(\cdot|x^*).$
- Consider the sequence of (random) posteriors $\Pi_n = p(\cdot|Y_{1:n})$ $n \to \infty$

Result: Π_n approaches a normal distribution.

- Which normal distribution?
- The same normal that the maximum likelihood estimator converges to!
    - Centered at x^*,
    - variance is the inverse of the observed information.

posterior

$$d(\Pi_n, \mathcal{N}(x^*, I_n^{-1})) \to 0, \text{ as } n \to \infty.$$

与 prior 无关.

MLE

Fisher Info

where:

- d is a distance between distributions called the total variation distance,
- the observed information I_n is given by

$$I_n = -\frac{\partial^2}{\partial x^2} \log p(Y_{1:n}|x) \Big|_{x^*},$$

- convergence is in probability, with respect to the distribution $p(\cdot|x^*)$.

Conditions include the following highlights:

1. the prior $p(x)$ is a density assumed to be non-zero in a neighborhood of the true parameter,
2. we assume the posterior mean is **consistent** as $n \to \infty$ (something you will explore in this week's exercise and lab),
3. we assume the model is **parametric**:
    - x is allowed to be a vector…
    - …but you are not allowed to make it longer as the number of observations increases.

References

- BDA, Section 2 and Appendix B (note however that they use a non-standard

definition of observed information).

- Full technical conditions in e.g., van der Vaart, A. W. (1998). Asymptotic Statistics. Cambridge University Press. ISBN 0-521-49603-9.

Laplace approximation

Outline

Topics

- Laplace: an approximation to the posterior motivated by BvM.
- Example of Laplace's failure modes.

Rationale

Laplace approximation is a useful tool for some (but not all) Bayesian models. The Laplace approximation is attractive because it is sometimes very cheap to compute and accurate. Understanding that approximation will also help understanding the BvM theorem introduced in the previous page.

Example

We use the same example as this week's lab:

$$X \sim \mathrm{Unif}(0,1)$$
$$Y|X \sim \mathrm{Binom}(n, X).$$

We have implemented $\gamma(x) = p(x, y)$ for this model in R, in a function called `lab5logdensity`.

▶ Code

Since we study the big data regime, we set the number of data points fairly large:

```
n_trials
```

```
[1] 800
```

The observed data y is:

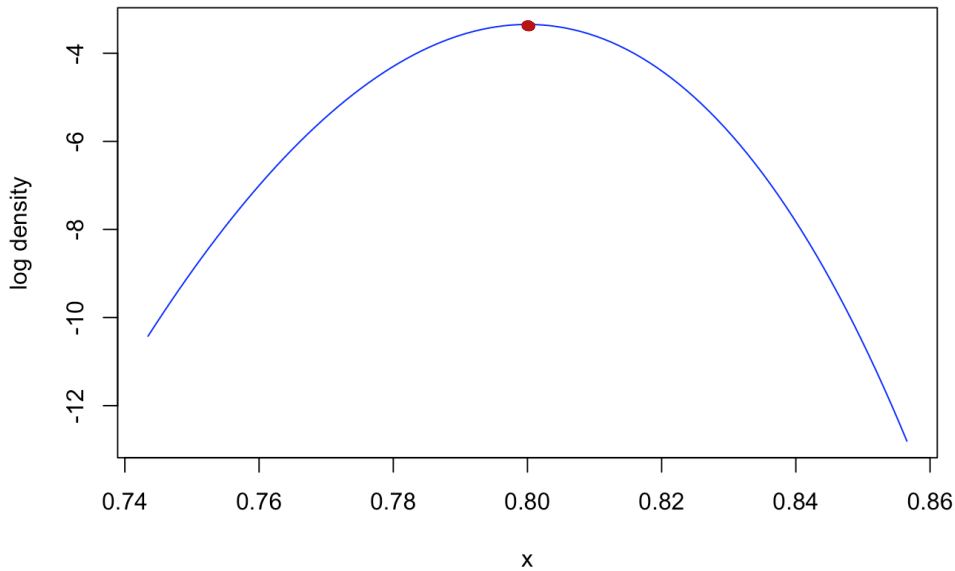
```
n_successes
```

```
[1] 640
```

Motivation

Let us look at the log density near the mode of the posterior

```
opt = maximize(lab5logdensity, 0.5)

second_derivative = opt$second_derivative
zoomed_width = 4 / sqrt(-second_derivative)
center = opt$argmax
left = center - zoomed_width
right = center + zoomed_width

curve(lab5logdensity, from = left, to = right, col = "b
```

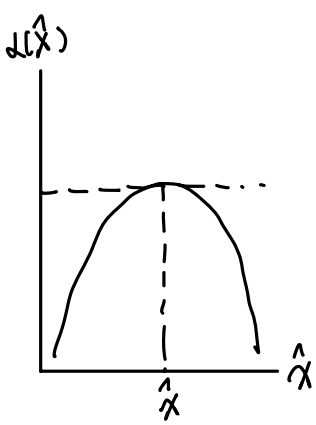


Question: How would you approximate such a function? quadratic function

⇒ Taylor expansion up to second degree and centered at mode.

Click for answer

Taylor expansion



Notation: denote the function above by $\ell(x) = \log \gamma(x)$.

Notation: denote the MAP (i.e., $\arg\max \ell$) by $\hat{x}$.

Recall Taylor's theorem (shown here for the second order):

$$\ell(x) = \ell(\hat{x}) + \ell'(\hat{x})(x - \hat{x}) + \frac{\ell''(\hat{x})}{2!}(x - \hat{x})^2 + R(x),$$

error.

quadratic expansion

where different flavours of Taylor's theorem will give different types of control of the residual error term $R(x)$ (we will not go into that level of detail today).

Question: one term in the Taylor expansion is zero. Which one? Why?

$\ell'(\hat{x}) = 0$ since $\hat{x}$ is the location of max

$\Rightarrow \ell(x) = \ell(\hat{x}) + \frac{\ell''(\hat{x})}{2!}(x - \hat{x})^2.$

Click for answer

Implemetation

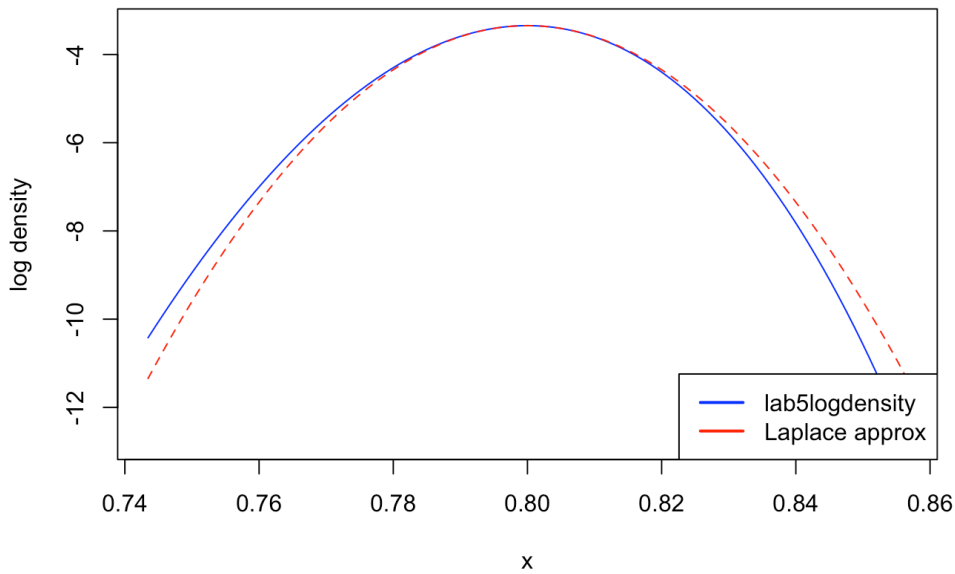
All the ingredients in the Taylor approximation can be obtained using an optimizer package. For the second derivative, either automatic differentiation or finite differences can be used.

Here is a simple implementation:

optimizer.

```
laplace_approx = function(opt) {

  x_hat = opt$argmax
  maximum = opt$max
  second_derivative = opt$second_derivative

  approx = function(x) {
    return(maximum + second_derivative / 2 * (x - x_hat
  }

  return(approx)
}

laplace = laplace_approx(opt)

curve(lab5logdensity, from = left, to = right, col = "b
curve(laplace, from = left, to = right, add = TRUE, col

legend(
  "bottomright",
  legend = c("lab5logdensity", "Laplace approx"),
  col = c("blue", "red"),
  lwd = 2
)
```

Taylor expansion.



As you can see, the approximation is quite good where it matters (near the mode)!

Let use denote the Laplace approximation by $\log q$:

$$\ell(x) \approx \underbrace{\ell(\hat{x}) + \frac{\ell''(\hat{x})}{2}(x - \hat{x})^2}_{\log q(x)}.$$

Question: can you interpret $q(x)$ as a density?

$q(x) = \exp(\ell(\hat{x}) + \frac{\ell''(x)}{2}(x-\hat{x})^2)$

$\propto \exp(\frac{\ell''(x)}{2}(x-\hat{x})^2)$

Click for answer

Hence we have to match the above with the parametrization of a normal, given by:

$$\underbrace{\text{constant}}_{\hat{x}} \times \exp\left(-\frac{(x-\mu)^2}{2\sigma}\right).$$

Hence we have $\mu = \hat{x}$ and $\sigma = -1/\ell''(\hat{x})$.

Asymptotic analysis of Laplace's approximation quality

- Consider the same setup as BvM (increasing random datasets).
- Denote the variance of the Laplace approximation computed from $Y_{1:n}$ by σ_n^2.
- Recall that the asymptotic normal distribution in BvM had variance $-I_n^{-1}$.

Note: $\lim_{n\to\infty} \sigma_n^2/(-I_n^{-1}) = 1$.

To see why, notice that the second derivative of ℓ is the sum of the prior's second derivative at $\hat{x}$, denoted $H = -(\log p(\cdot))''(\hat{x})$, and the observed information I_n, hence

$$\lim \frac{\sigma_n^2}{-I_n^{-1}} = \lim \frac{I_n}{H + I_n} = \lim \frac{\frac{1}{n} I_n}{\frac{1}{n} H + \frac{1}{n} I_n}.$$

Finally, $\frac{1}{n}H$ converges to zero, while by the law of large numbers, $\frac{1}{n}I_n$ converges to a constant, namely the Fisher information, $\mathbb{E}[I_n]$.

Conclusion:

- The Laplace approximation converges to the same normal as the one appearing in BvM.
- Hence, when BvM applies, the Laplace approximation also gets arbitrarily accurate as $n \to \infty$.

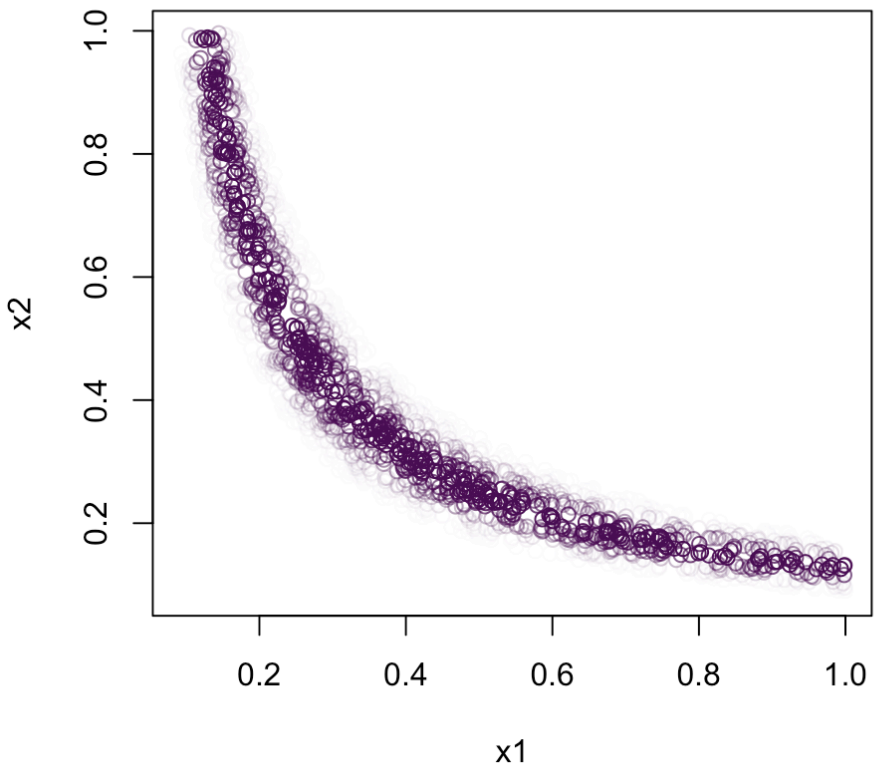
Typical failure mode

Consider the following model:

$$\begin{aligned} X_1 &\sim \mathrm{Unif}(0,1) \\ X_2 &\sim \mathrm{Unif}(0,1) \\ X &= X_1 X_2 \\ Y|X_1, X_2 &\sim \mathrm{Binom}(n, X). \end{aligned}$$

Let us use simPPLe to approximate the posterior given a fairly large number of observations ($n = 800$):

```
source("../../solutions/simple.R")
source("../blocks/simple_utils.R")

unid_size = 800

unidentifiable = function() {
  x1 = simulate(Unif(0, 1))
  x2 = simulate(Unif(0, 1))
  x = x1 * x2
  observe(unid_size / 8, Binom(unid_size, x))
  return(c(x1, x2))
}

posterior = posterior_particles(unidentifiable, 10000)
weighted_scatter_plot(posterior, plot_options = list(xl
```



- In 2 dimensions, the Laplace approximation is a bivariate normal.
- Hence the contour lines of Laplace approximations are restricted to ellipses.
- This means that the Laplace approximation would be very poor here.

Question: which of BvM's key assumptions is violated here?